

PEG crosslinked poly(vinylbenzene boronic acid) polymer electrolytes for Li-ion batteries



Sevim Ünügür Çelik, Ayhan Bozkurt*

Department of Chemistry, Fatih University, 34500 Büyükdere, Istanbul, Turkey

ARTICLE INFO

Article history:

Received 18 April 2012

Received in revised form

12 June 2013

Accepted 13 June 2013

Available online 24 June 2013

Keywords:

Poly(4-vinylbenzenboronic acid)

Polymer electrolyte

Li-ion conductivity

PEG

ABSTRACT

Poly(4-vinylbenzenboronic acid), PVBBA was synthesized via free-radical polymerization of 4-vinylbenzenboronic acid (4-VBBA) and followed by crosslinking with polyethylene glycol (PEG) with different molecular weights to produce boron containing crosslinked polymers. Prior to crosslinking, the materials were doped with $\text{CF}_3\text{SO}_3\text{Li}$ at several stoichiometric ratios to get PVBBAPEGX-Y where X is the molecular weight of PEG and Y is the EO/Li ratio. The materials were characterized by using Fourier transform infrared spectroscopy (FT-IR), thermogravimetric analysis (TGA) and differential scanning calorimeter (DSC). The ionic conductivity of these novel crosslinked electrolytes was studied by dielectric-impedance spectroscopy. Li-ion conductivity of these polymer electrolytes depends on the length of the side units as well as the doping ratio. PVBBAPEG200-10 illustrated a satisfactory ionic conductivity of $3.1 \times 10^{-5} \text{ S/cm}$ at 20°C and $1.8 \times 10^{-3} \text{ S/cm}$ at 100°C .

© 2013 Elsevier B.V. All rights reserved.

1. Introduction

Polymer electrolytes that have the ability to transport lithium ions are important for both technological and scientific areas due to their potential applications in rechargeable lithium-ion batteries [1,2]. In generally, polymer electrolytes can be prepared by mixing of salt (LiX) in a suitable polymer matrix where there is an interaction between ethylene oxide units of the polymer and guest molecules [3–5]. The electrolytes based on polyethylene oxide (PEO) complexed with alkali metal salts are the most extensively studied systems [6–8]. Depending on the salt concentration these polymer electrolytes may have semicrystalline morphology which may negatively affect the ionic conductivity especially at lower temperatures.

Moreover, polymer electrolytes can also be prepared by mixing of a stable polymer matrix with additives such as ethylene carbonate (EC), propylene carbonate (PC), dimethyl carbonate (DMC) and N-methylpyrrolidinone (NMP) (BL) to promote ionic conductivity [9–11]. Polymers such as polyvinylidene fluoride (PVDF), polyacrylonitrile (PAN), poly(methyl methacrylate) (PMMA) and polyvinyl chloride (PVC) have been used as host materials [12–14]. These systems comprise functional groups that are not sufficient to create charge separation between electrolyte ions. Thus, special

additives are required to promote sufficient charge separation, which enable the ions in the solid matrix to respond to an electric field. In these systems, the major role of the polymer is mostly to maintain a solid and stable matrix, whereas the ion migration within the matrix under an electrical field is feasible because of the additives. However, the mechanical property of such polymer electrolytes is generally poor because of the characteristic of gels and additives [15].

Moreover, a variety of grafted polymers was investigated, in which PEO side chains were attached to a polymer backbone, which itself had a low glass transition temperature and thus giving rise to a more flexible system [16–19]. In another previous example grafted systems which based on polyphosphazene (MEEP) backbones and PEO side chains were investigated. These materials are completely amorphous with glass transition temperatures around -80°C . However, the room-temperature conductivities of the $\text{F}_3\text{CSO}_3\text{Li}$ salt mixtures close to 10^{-5} S/cm [20].

Poly(4-vinylbenzenboronic acid), PVBBA is a thermally and chemically stable polymer and used as host matrix for proton conducting membrane [21]. It is an attractive polymeric support for polymer-supported esterification reaction with diols [22]. The functionalization can easily be done by the attachment of hydroxyl groups in a suitable solvent.

In this work, poly(4-vinylbenzenboronic acid), PVBBA was synthesized via free-radical polymerization of 4-vinylbenzenboronic acid (4-VBBA) and followed by crosslinking with polyethylene glycol (PEG) with different molecular weights to produce

* Corresponding author. Tel.: +90 212 8663300.

E-mail address: bozkurt@fatih.edu.tr (A. Bozkurt).

boron containing PEG crosslinked polymers. Prior to crosslinking, the doping with $\text{CF}_3\text{SO}_3\text{Li}$ was achieved at several EO/Li ratios. The polymer electrolytes were characterized via FT-IR, TGA and DSC. Li-ion conducting properties of the materials were investigated by dielectric-impedance analyzer and the results are discussed and compared with previously reported systems.

2. Materials

4-Vinylbenzeneboronic acid (>98%) was provided from Alfa Aesar and the initiator α,α' -azodiisobutyramidine dihydrochloride (AIBADC) was supplied from Aldrich. Poly(ethylene glycol) (PEG, average $M_n = 200, 600$ and 1000) were purchased from Aldrich. Dimethyl sulfoxide (DMSO, analytical reagent) was obtained from Merck. PEG and DMSO were dried by using molecular sieves. $\text{CF}_3\text{SO}_3\text{Li}$ was received from Aldrich and stored in the glove box.

2.1. Sample preparation

Poly(4-vinylbenzeneboronic acid), PVBBA, was produced by free-radical polymerization of 4-vinylbenzeneboronic acid in DMSO using AIBADC as initiator according to our previous work [21]. The polymerization solution was purged with nitrogen and the reaction was performed in DMSO at 85°C for 4 h. After polymerization, the resulting solid was filtered and washed several times with excess DMSO and then dried under vacuum at 50°C . A mixture of PVBBA and PEG (stoichiometric molar ratio = 1:1) was refluxed in glove box at 100°C and 48 h (Fig. 1). Then polymer gels were washed in excess diethylether, at least three times, to remove unreacted PEG. Then the materials were dried under vacuum at 60°C to get crosslinked polymers with three different PEG spacers, i.e., PVBAPEG200, PVBAPEG600 and PVBAPEG1000.

Since the crosslinked polymers are not soluble in any solvent, the polymer electrolytes were prepared by adding of appropriate amounts of $\text{CF}_3\text{SO}_3\text{Li}$ salt prior to crosslinking of PVBBA with PEG. The concentration of Li-salt was adjusted so that the molar ratio of lithium atoms to ether oxygen atoms in the total PEG of

Table 1

The composition and properties of PVBAPEGX-Y samples.

Sample PVBAPEGX-Y	Y (EO:Li mol ratio)	T_d ($^\circ\text{C}$)	T_m ($^\circ\text{C}$)	Max. ion conductivity (S/cm) at 100°C
PVBAPEG200	—	200	—	—
PVBAPEG200-10	10:1	200	—	1.85×10^{-3}
PVBAPEG200-20	20:1	200	—	3.30×10^{-4}
PVBAPEG200-30	30:1	200	—	6.15×10^{-4}
PVBAPEG600	—	300	10	—
PVBAPEG600-10	10:1	300	—	1.92×10^{-3}
PVBAPEG600-20	20:1	300	—	3.34×10^{-4}
PVBAPEG600-30	30:1	300	10	5.49×10^{-4}
PVBAPEG1000	—	350	40	—
PVBAPEG1000-10	10:1	350	—	1.94×10^{-3}
PVBAPEG1000-20	20:1	350	30	8.90×10^{-4}
PVBAPEG1000-30	30:1	350	30	6.21×10^{-4}

PVBAPEG200, PVBAPEG600 and PVBAPEG1000 (Table 1). The samples were abbreviated as PVBAPEGX-Y where X denotes the molecular weight of the PEG and Y is the doping ratio, i.e., EO/Li ($Y = 10, 20$ and 30). The resulting homogeneous sticky materials were dried at 60°C and stored in glove box.

2.2. Characterizations

FT-IR measurements were performed using Bruker Alpha ATR ($4000\text{--}400\text{ cm}^{-1}$, resolution 4 cm^{-1}).

Thermal stabilities of the polymer electrolytes were examined by TG analyses with a Perkin Elmer STA 6000 Thermal Analyzer. The samples ($\sim 10\text{ mg}$) were heated from room temperature to 700°C under N_2 atmosphere at a scanning rate of $10^\circ\text{C}/\text{min}$.

DSC measurements were carried out on a Perkin Elmer Jade DSC under nitrogen atmosphere and heating–cooling curves were recorded at a rate of $10^\circ\text{C}/\text{min}$. The melt temperature (T_m) was evaluated from the onset temperature of melting.

The real (ϵ') and imaginary (ϵ'') parts of complex dielectric permittivity ($\epsilon^* = \epsilon' - i\epsilon''$) were measured with a Novocontrol dielectric-impedance analyzer. The films were sandwiched

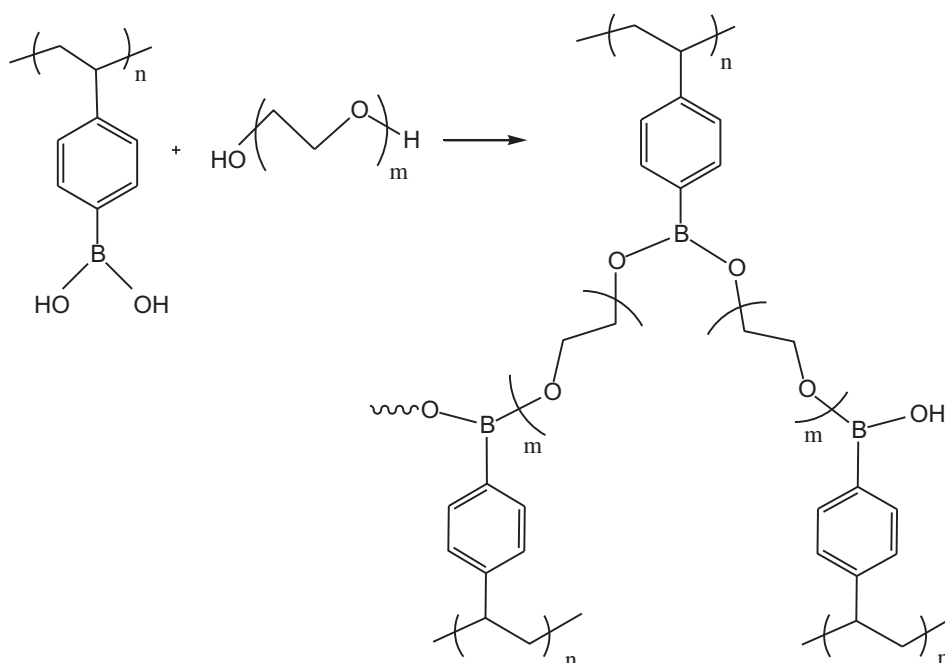


Fig. 1. Synthesis scheme of crosslinked polymer electrolytes.

between platinum blocking electrodes and the conductivities were measured in the frequency range 0.1 Hz–3 MHz at 10 °C intervals. The temperature was controlled with a Novocontrol Cryosystem, which is applicable between –100 °C and 250 °C. The dielectric data (ϵ' , ϵ'') were collected during heating as a function of frequency. For conductivity data analysis, the real part of conductivity (σ') was calculated from the complex permittivity using the relation, $\sigma'(\omega) = \sigma_{ac}(\omega) = \epsilon''(\omega) \omega \epsilon_0$, where ω is the angular frequency and ϵ_0 is the vacuum permittivity.

3. Results and discussion

3.1. FT-IR study

FT-IR spectra of PVBBAPEG600 and PVBBAPEG1000 series are illustrated in Fig. 2a and b, respectively. Both series show similar characteristic of PVBBA and PEG peaks. The absorption bands at 2870 cm^{-1} are assigned to the symmetric and antisymmetric stretching vibration of aliphatic CH units. Two strong peaks at 1722 cm^{-1} and 1610 cm^{-1} are attributed to C=C stretching bands of

p-disubstituted benzene ring. The strong CH out-of plane deformation band for p-disubstituted benzene ring is at 830 cm^{-1} . The vibration of phenyl boronic acid linkage (Ph–B) gives a strong peak at 1458 cm^{-1} . The absorption band at 1352 cm^{-1} is assigned to the vibration of the bond B–O. The broad absorption band around 3400 cm^{-1} that correspond to absorption of –B(OH) groups, was very weak, which illustrated high conversion of the hydroxyl groups. The strong peaks at 1085 cm^{-1} may be attributed to C–O–C units of the PEG chains. The peak at 1030 cm^{-1} can be attributed to stretching vibration of SO_3 unit of the dopant and increases with the dopant ratio.

3.2. Thermal analysis

Thermal stability is an important property for polymer electrolytes during application in lithium-ion polymer batteries. The thermal stability was studied by TGA for the PVBBA, crosslinked polymers and the Li-salt doped polymers. Table 1 and Fig. 3 show the TGA results of the polymer electrolytes. Pure PVBBA has a considerable weight loss up to 400 °C due to condensation of boronic acid units. After grafting with PEG, this region is not observed in TGA curves which confirms the grafting of B(OH)_2 units. For all the polymer electrolytes no weight loss were observed until 200 °C and the thermal stability increased with molecular weight of PEG. PVBBAPEG200-20 is thermally stable up to 200 °C and then 15% of the sample decomposes until 350 °C. The T_d temperatures are shifted to 300 °C for PVBBAPEG600 series and 350 °C for PVBBAPEG1000 series. For all the samples the major weight loss occurs after 350 °C and the minor weight loss before the 300 °C may be due to condensation of unreacted B(OH) groups. For all the samples there is not weight loss before 100 °C which confirm that complete dryness of the materials.

Fig. 4(a and b) shows the melt endotherms of semi-crystalline crosslinked polymer and the doped samples. For PVBBAPEG200 series there is not a definite T_g or melting point. The polymer samples PVBBPEG600 and PVBBAPEG1000 in the absence of dopant illustrate melting temperatures 10 °C and 40 °C, respectively. The sample PVBBPEG600-30 displays similar T_m . The melting temperatures of PVBBPEG1000-20 and PVBBPEG1000-30 are around 30 °C. Clearly, the material showed more crystalline behavior with increasing PEG molecular weight and the addition of more dopant

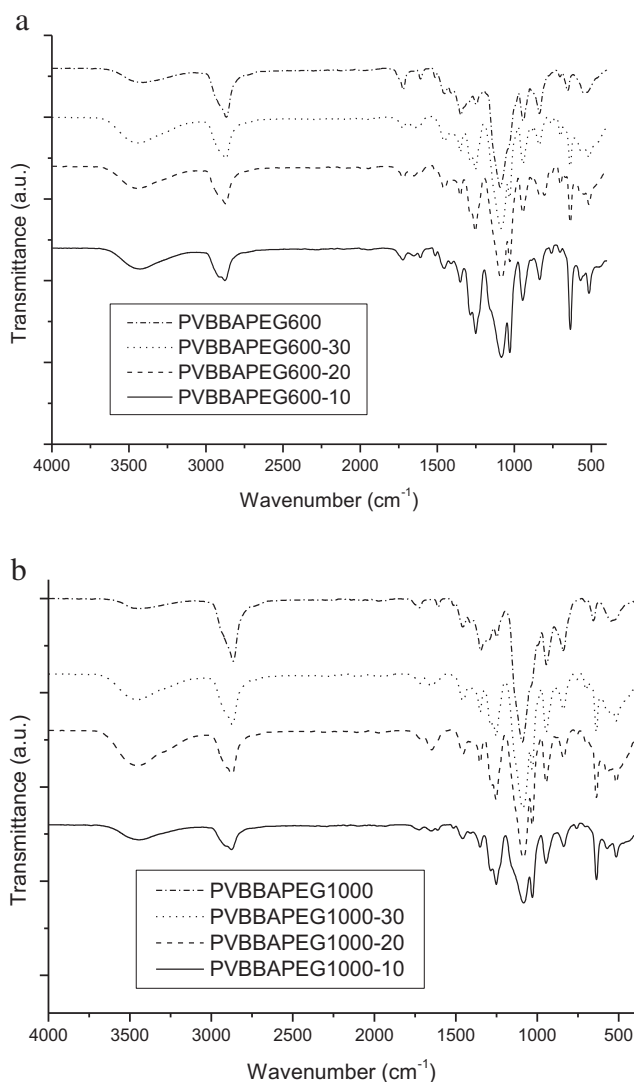


Fig. 2. (a) FT-IR spectra of PVBBAPEG600 and PVBBAPEG600-Y (Y = 10, 20, 30) polymer electrolytes. (b) FT-IR spectra of PVBBAPEG1000 and PVBBAPEG1000-Y (Y = 10, 20, 30) polymer electrolytes.

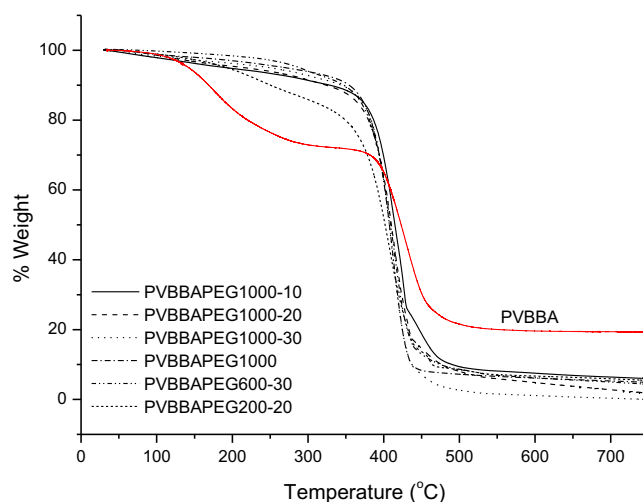


Fig. 3. Comparison of the weight loss of PVBBA, PVBBAPEG200-20, PVBBAPEG600-30, PVBBAPEG1000 and PVBBAPEG1000-Y (Y = 10, 20, 30) polymer electrolytes. The heating rate was 10 °C/min.

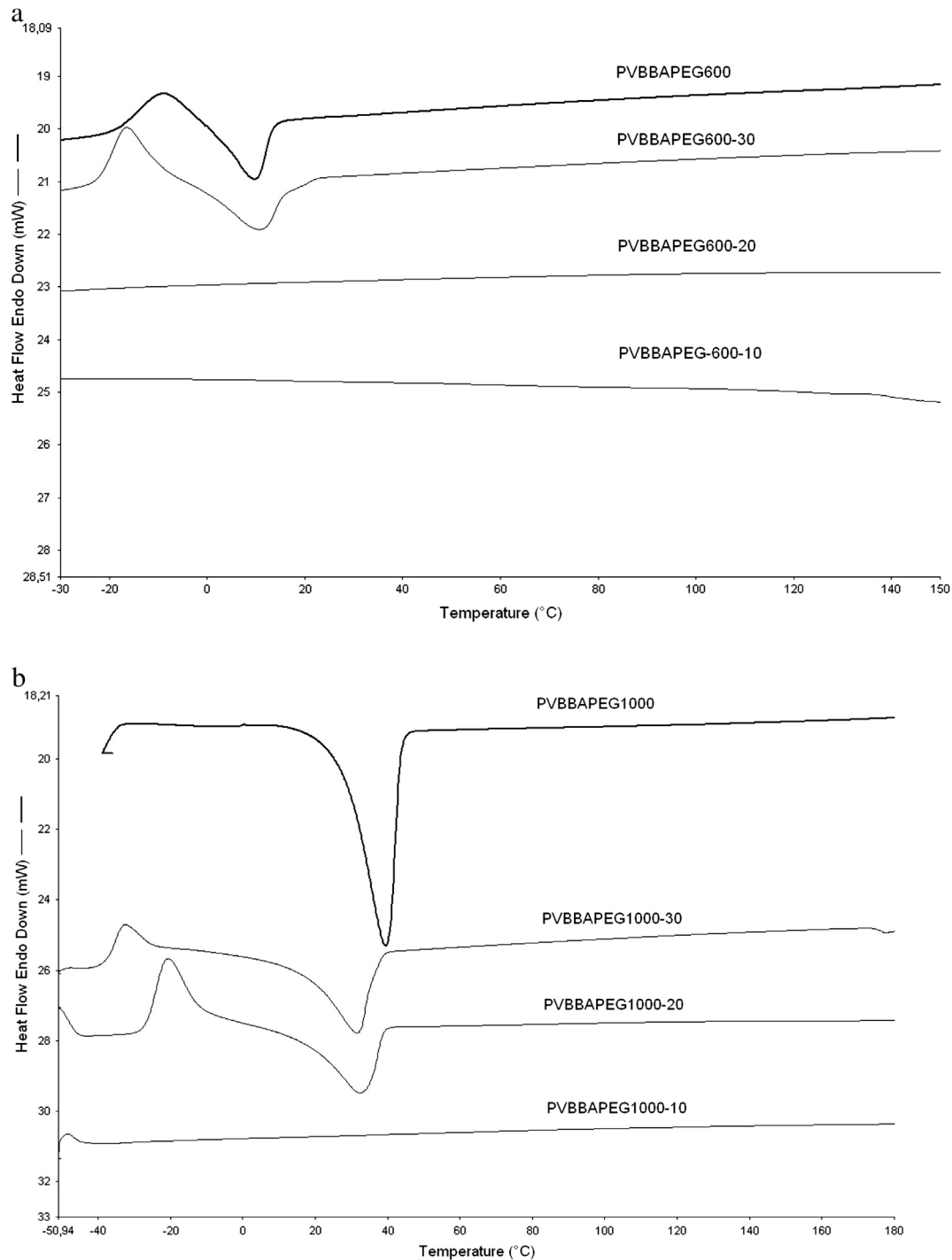


Fig. 4. (a) DSC curves of PVBBAPEG600 and PVBBAPEG600-Y (Y = 10, 20, 30) polymer electrolytes. Second heating run endotherms are presented at a heating rate of 10 °C/min. (b) DSC curves of PVBBAPEG1000 and PVBBAPEG1000-Y (Y = 10, 20, 30) polymer electrolytes. Second heating run endotherms are presented at a heating rate of 10 °C/min.

results more amorphous character. The presence of single T_g verifies the homogeneity of the materials.

3.3. Conductivity measurement

The ionic conductivity of lithium ion in polymer electrolytes containing PVBBAPEGX-Y with changing PEG into chain spacers was measured by the ac impedance technique. The samples were prepared as thin films in the glove box under nitrogen atmosphere with humidity <0.1 ppm. The conductivity measurements have also been

carried out in a completely water-free environment in order to eliminate the contribution of humidity to the ion conductivity. The samples were cut into disks of 10 mm in diameter sandwiched between Pt electrodes and subjected to the impedance analyzer.

Frequency dependence of ac conductivities of PVBBPEG600-10 and PVBBPEG1000-20 at various temperatures are shown in Fig. 5a and b. Frequency dependent ac conductivities ($\sigma_{ac}(\omega)$) were measured using Eq. (1):

$$\sigma'(\omega) = \sigma_{ac}(\omega) = \varepsilon''(\omega)\omega\varepsilon_0 \quad (1)$$

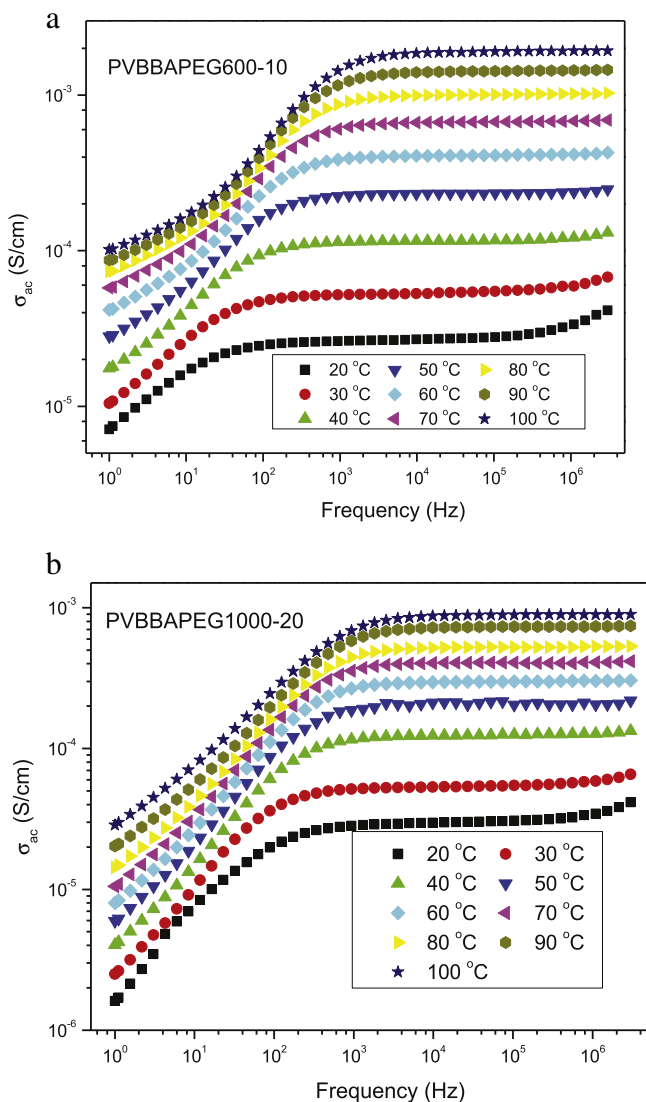


Fig. 5. (a) AC conductivity versus frequency of anhydrous PVBBAPEG600-10 at various temperatures. (b) AC conductivity versus frequency of anhydrous PVBBAPEG1000-20 at various temperatures.

where $\sigma'(\omega)$ is the real part of conductivity, $\omega = 2\pi f$ is the angular frequency, ϵ_0 is the vacuum permittivity ($\epsilon_0 = 8.852 \times 10^{-14} \text{ F cm}^{-1}$), and ϵ'' is the imaginary part of complex dielectric permittivity (ϵ^*). As seen in Fig. 5a and b, the frequency dependent ac conductivity curves show a general trend for all samples and at all temperatures in the log–log plot. It can be distinguished clearly from the figures that the curves of ac conductivity versus frequency for various temperatures exhibits frequency dependent linear increase at low frequency regime which is attributed to the polarization of the blocking electrodes. Additionally, it is strongly temperature dependent within this frequency zone and has the ac conductivity in the order of 3.7×10^{-6} – 5.2×10^{-4} for PVBBAPEG600-10 and 8.5×10^{-6} – 1.6×10^{-5} for PVBBAPEG1000-30 (2.7×10^{-6} – 5.5×10^{-4} for PVBBAPEG200-10 and 3.3×10^{-6} – 1.1×10^{-5} PVBBAPEG1000-20). A well-developed frequency independent plateau is observed at medium frequencies. The frequency independent conductivity is, in general, identified with the dc conductivity (σ_{dc}) which was obtained from the plateau region in graphs of σ_{ac} vs Freq. by linear fittings. The reasonable model for the lithium ion conduction in this crosslinked polymer system is

considered to be similar to the previously studied host matrix system [23]. However, the flexibility of the side chains is limited in this new system since the polymer has branched stiff boron containing PVBBA chains as reinforcing backbone to provide a network structure. In this network system, lithium trifluoromethane sulfonate is dissociated into ions, which are dispersed and stabilized by oxygen atoms of ethylene oxide units in the polymer chains. Cations tend to move toward the cathode while anions move in the opposite direction upon applied potential. At the same time, the charges on the interface of electrode and electrolyte will rearrange.

DC conductivity plots versus reciprocal temperature for all the polymer/salt complexes are illustrated in Fig. 6. The curved lines indicates that the ion conduction in these complexes can be explained the concept of free volume [24,25], which is expressed by the Vogel–Tamman–Fulcher (VTF) equation (Eq. (2)).

$$\sigma = \sigma_0 \exp(-B/k(T - T_0)) \quad (2)$$

where σ_0 is the prefactor, T is the absolute temperature, k is the Boltzmann constant, B is an apparent activation energy and T_0 is the equilibrium glass transition temperature at which the “free” volume disappears or at which configuration free entropy becomes zero (i.e., molecular motions cease).

A comparison between the conductivities of the electrolytes indicates that the length of the PEG main chain segment does not have significant influence on the conductivity as a function of temperature since the all electrolytes shows comparable conductivities with the same salt concentration. Previously, it was shown that the conductivities of the electrolytes were increasing with the length of the PEG incorporated in the polymer main chain at a given temperature due to the plasticization effect of PEG [23]. This was attributed to a decrease of the T_g 's of the different electrolytes, and thus an increasing of the mobility of the polymer chains. In the present study, however, a given salt concentration, the conductivity was found to be independent of the T_g 's of the electrolytes since the electrolytes show no T_g transition in the investigated temperature range. It may also confirm that PEG does not show a considerable plasticizing effect as reported in our previously work [26]. Despite this fact that the measured conductivities of these non-plasticized polymer electrolytes was found to be exceptionally high where it is $1.85 \times 10^{-3} \text{ S/cm}$ at 100 °C and $3.65 \times 10^{-5} \text{ S/cm}$ at 20 °C for PVBBAPEG200-10. Such an ionic conductivity can be described by

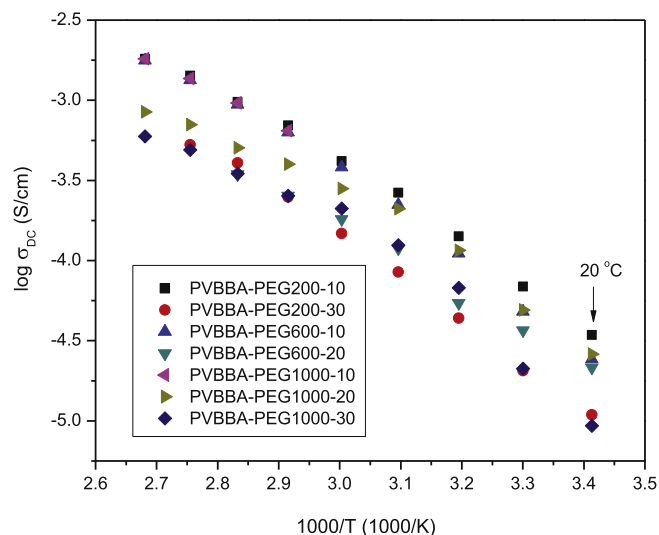


Fig. 6. Temperature dependence of the DC conductivities of PVBBAPEGX-Y.

molecular structure of the host polymer. Clearly, faster lithium ion conduction occurs in the polymer matrix by the fast conduction pathways created by highly ordered arrangement of Li^+ co-ordination EO sites along the PVBBA backbone.

The variation of conductivity at different temperatures as a function of EO/Li ratio is shown in Table 1. The conductivity increased regularly with increasing salt concentration reaching a maximum for EO/Li = 10:1 (i.e., $\sim 1 \times 10^{-3}$ S/cm at 80 °C, and $\sim 1.9 \times 10^{-3}$ S/cm at 100 °C). The further addition of salt decreases the conductivity. This behavior has previously been attributed to the formation of ionic clusters, reducing the number and the overall mobility of the ions [27]. In general, the ionic aggregates do not contribute significantly to the conductivity values as their comparatively large radii considerably reduce their mobility in electrolytes [28]. However, increasing temperatures normally lead to a higher degree of salt dissociation [29]. Thus, the number of charge carriers is increased which enhances the conductivity. The reduction of the conductivity with increasing salt concentration may also be explained by the highly crosslinked structure of present polymer, with Lewis acidic boron atoms at the branch points, which may provide the EO units closest to the branch points ineffective as coordinating sites for Li^+ . Additionally, the interactions between Li^+ and EO units are, in general, stronger than the interactions between the anion and the Lewis acid, particularly if the anion has a “soft” character. The incorporation of boronate ester in the architecture of the polymer both reduced the concentration of EO units and the level of interactions between the cation and the polymer. Mizumo et al. studied a polymer based only on Lewis acidic borane and alkyl units [30]. It has been showed that the solubility of salts in this polymer was very poor as compared to the solubility observed in electrolytes containing only Lewis basic EO units. Consequently, a higher level of trapping of the CF_3SO_3^- anion by the boron, connected with a reduction of ionic mobility, should be more relevant as an explanation to the decrease of the conductivity.

4. Conclusions

In the present work, a novel crosslinked boron containing polymers with different PEG side chains were produced. Then polymer electrolytes were synthesized after doping the host material with lithium salt at several molar ratios, i.e., EO/Li. The PEG spacers in polymer is capable of forming donor:acceptor type bonds by complexing with the metal ion. These polymer

electrolytes were characterized using DSC, FT-IR, TGA and dielectric-impedance spectroscopy. The conductivity studies illustrate that the ionic conductivity of these polymer electrolytes was found to depend on side chain length as well as the Li^+ content. The polymer electrolyte PVBBAPEG200-10 illustrated a satisfactory ionic conductivity of 3.1×10^{-5} S/cm at 20 °C and 1.8×10^{-3} S/cm at 100 °C.

Acknowledgments

This work was supported by BOREN under contract number 2009.Ç0213.

References

- [1] A.M. Spethan, K.S. Nahm, *Polymer* 47 (2006) 5952.
- [2] J.M. Tarascon, M. Armand, *Nature* 414 (2001) 359.
- [3] K.M. Abraham, *Electrochim. Acta* 38 (1993) 1233.
- [4] P.G. Bruce, *Electrochim. Acta* 40 (1995) 2077.
- [5] Z. Gadajourova, Y.G. Andreev, D.P. Tunstall, P.G. Bruce, *Nature* 412 (2001) 520.
- [6] P. Lightfoot, M.A. Metha, P.G. Bruce, *Science* 262 (1993) 883.
- [7] D.E. Fenton, J.M. Parker, P.V. Wright, *Polymer* 14 (1973) 589.
- [8] M.B. Armand, S.M. Chabagno, M. Duclot, in: *Second International Meeting on Solid Electrolytes*, St. Andrews, Scotland, Extended Abstracts, 20–22 Sept., 1978 (1978).
- [9] Y.K. Yarovoy, H.P. Wang, S.L. Wunder, *Solid State Ionics* 118 (1999) 301.
- [10] H.P. Wang, H.S. Huang, L. Wunder, *J. Electrochem. Soc.* 147 (2000) 2853.
- [11] K. Murata, S. Izuchi, Y. Yoshihisa, *Electrochim. Acta* 45 (2000) 1501.
- [12] Y.J. Shen, M.J. Reddy, P.P. Chu, *Solid State Ionics* 175 (2004) 747.
- [13] Q. Xiao, X. Wang, W. Li, Z. Li, T. Zhang, H. Zhang, *J. Membr. Sci.* 334 (2009) 117.
- [14] D.J. Lin, C.L. Chang, C.K. Lee, L.P. Cheng, *Eur. Polym. J.* 42 (2006) 2407.
- [15] M. Wang, L. Qi, F. Zhao, S. Dong, *J. Power Sources* 139 (2005) 223.
- [16] W.H. Meyer, *Adv. Mater.* 10 (1998) 439.
- [17] P.G. Hall, G.R. Davis, J.E. McIntyre, I.M. Ward, D.J. Banister, K.M.F. Le Brocq, *Polym. Commun.* 27 (1986) 98.
- [18] D. Fish, I.M. Khan, J. Smid, *Macromol. Chem. Rapid Commun.* 7 (1986) 115.
- [19] R. Spindler, D.F. Shriver, *J. Am. Chem. Soc.* 10 (1988) 3036.
- [20] P.M. Blonsky, D.F. Shriver, P. Austin, H.R. Allcock, *J. Am. Chem. Soc.* 106 (1984) 6854.
- [21] A. Sezgin, Ü. Akbey, M.R. Hansen, R. Graf, A. Bozkurt, A. Baykal, *Polymer* 49 (2008) 3859.
- [22] G. Belogi, T. Zhu, G.J. Boons, *Tetrahedron Lett.* 41 (2000) 6965.
- [23] S.Ü. Çelik, A. Bozkurt, *Solid State Ionics* 181 (2010) 987–993.
- [24] G.S. Grest, M.H. Cohen, *Phys. Rev. B* 21 (1980) 4113.
- [25] Y. Kato, K. Hasumi, S. Yokoyama, T. Yabe, H. Ikuta, Y. Uchimoto, M. Wakihara, *Solid State Ionics* 150 (2002) 355.
- [26] H. Aydın, A. Bozkurt, *J. Appl. Polym. Sci.* 124 (2012) 1193–1199.
- [27] R.G. Linford, *Application of Electroactive Polymers*, first ed., Chapman & Hall, London, 1993.
- [28] S.J. Wen, T.J. Richardson, D.I. Gbantous, K.A. Striebel, P.N. Ross, E.J. Cairns, *J. Electroanal. Chem.* 408 (1996) 113.
- [29] I. Rey, J.C. Lasségues, J. Grondin, L. Servant, *Electrochim. Acta* 43 (1998) 1505.
- [30] N. Matsumi, T. Mizumo, H. Ohno, *Chem. Lett.* 33 (2004) 372.